

A published standalone section (Tag 0SE01)

Definition 1 (Readable typography) (Tag 0SE0101). A template is readable when its visual hierarchy supports the mathematics without competing with it.

The local references are [Section 1 \(Tag 0SE01\)](#) and [Definition 1 \(Tag 0SE0101\)](#).

1 A tagged subsection (Tag 0SE0102)

Section mode uses a centered page number and no header or footer rules.

A draft environment gallery (Draft tag 0SE02)

1 Every statement style (Draft tag 0SE0201)

Slogan *Short independently compiled sections should still share the book's voice.*

Theorem A (Standalone coherence). A standalone section can use the same source as its chapter and book builds.

Definition 2. A chain complex is a sequence (C_n, d_n) with $d_{n-1} \circ d_n = 0$.

Proposition 3 (Cycles modulo boundaries). The quotient $\ker d_n / \operatorname{im} d_{n+1}$ is an abelian group.

Theorem 4. A short exact sequence of complexes induces a long exact sequence in homology.

Lemma 5 (Connecting map). The connecting morphism is independent of the chosen lift.

Corollary 6. A quasi-isomorphism induces isomorphisms on every homology group.

Conjecture 7. An analogous statement holds in the intended derived enhancement.

Question 8 (Naturality). How does the connecting map vary with a morphism of short exact sequences?

Example 9 (Draft tag 0SE0202). The complex concentrated in degree zero recovers an ordinary module. Its displayed draft tag does not appear in [Example 9](#).

Exercise 10. Write down the connecting map and verify that it is a homomorphism.

Construction 11 (Mapping cone). For $f : C_\bullet \rightarrow D_\bullet$, set $\operatorname{Cone}(f)_n = D_n \oplus C_{n-1}$ with the usual differential.

Notation 12. Write $H_n(C_\bullet)$ for the n -th homology group.

Remark 13 (Shared numbering). Every ordinary theorem-like environment participates in one local sequence.

Proof sketch of Theorem 4. Lift a cycle, apply the differential, and descend the resulting boundary.

Step 1 (Choose a lift). Select a representative in the middle complex.

Claim 14. Changing the lift changes the result by a boundary.

Case 1 (The class is zero). Choose a boundary representative and compute directly.

Case 2 (The class is arbitrary). Subtract two lifts and reduce to the first case.

□

References include Theorem A, Theorem 4, Lemma 5, Claim 14, Step 1, and Case 1. Draft structural references such as Section 2 and Subsection 1 omit tags.

2 Mathematics and code (Draft tag 0SE0203)

Inline mathematics remains calm in prose: $\mathcal{L} \otimes \mathcal{F} \in D^b(X)$ and $\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \mathbb{A}_k^n$. The same notation works in a labeled display:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots. \quad (1)$$

$$\begin{array}{ccccccc} 0 & \longrightarrow & A. & \longrightarrow & B. & \longrightarrow & C. \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A'. & \longrightarrow & B'. & \longrightarrow & C'. \longrightarrow 0 \end{array}$$

```
1 def homology(cycles, boundaries):
2     return cycles.quotient(boundaries)
```

Listing 1: A tiny section-level listing

See eq. (1) and Code 1.